

K-map

$$f = x\bar{y}(z+\bar{z}) + x\bar{y}z + x\bar{y}\bar{z} + x\bar{y}(z+\bar{z}) + x\bar{y}z + x\bar{y}\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + x\bar{y}z + x\bar{y}\bar{z}$$

$$m_1 + m_0 + m_7 + m_6 + m_2 + m_3 + m_4 + m_5$$

$$\Sigma m(1, 0, 7, 6, 2, 3, 4, 5)$$

	yz 00	01	11	10
x 0	1	1	1	1
1	0	0	0	1

$$y = \bar{x} + y\bar{z}$$

$$f = (x+y)(\bar{x}+y+z)(x+\bar{y}+\bar{z})$$

$$(x+y+z\bar{z})(\bar{x}+y+z)(x+\bar{y}+\bar{z})$$

$$(x+y+z)(x+y+\bar{z})$$

$$f(w,x,y,z) = \Sigma m(0, 3, 5, 6, 7) + D(1, 12, 13, 15)$$

wx \ yz	00	01	11	10
00	1	X	1	
01		1	1	1
11	X	X	X	0
10	1			

$$y = \bar{w}\bar{x}\bar{y} + \bar{w}z + xz + wx$$

Q) $AB + \bar{A}CB + A\bar{C}\bar{D}$

$A(B)$

$$AB(C + \bar{C})(D + \bar{D}) + \bar{A}CB(D + \bar{D}) + A\bar{C}\bar{D}(B + \bar{B})$$

$$ABCD + AB\bar{C}\bar{D} + ABC\bar{D} + AB\bar{C}D + \bar{A}CB\bar{D} + \bar{A}CB\bar{D} + A\bar{C}\bar{D}B + A\bar{C}\bar{D}\bar{B}$$

$\Sigma(m_{15} + m_{12} + m_{14} + m_{13} + m_7 + m_6 + m_{10})$

AB \ CD	00	01	11	10
00				
01			1	1
11	1	1	1	1
10				1

$F = A\bar{B} + B\bar{D} + A\bar{D}$

$\bar{F} = ?$

$\bar{F} = (\bar{A} + B)(\bar{B} + D)(A + \bar{D})$

$$F = A + \bar{B}$$

$$\bar{F} = \bar{A}B$$

A	B	C	F _a	F _b
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0

		F _a			
A	BC	00	01	11	10
	0		1		1
	1			1	1

$$F_a = \bar{A}\bar{B}C + AB + B\bar{C}$$

		F _b			
A	BC	00	01	11	10
	0		1	1	
	1		1		1

$$F_b = \bar{A}C + \bar{B}C + AB\bar{C}$$

$$F_a = \bar{A}(B \oplus C) + AB$$

Half adder to NOT gate

